Peter Leo

GEN AI FULL STACK DEVELOPER

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SUMMARY

AI/ML Systems Architect and Document Intelligence Engineer specializing in multimodal Vision-Language Models (VLMs), Generative AI, enterprise Document AI, computer vision, LLM/SLM fine-tuning, model quantization and high-throughput production microservices. Architected, built and deployed 15 enterprise AI systems across insurance, financial services, identity verification, claims automation, underwriting, fraud detection and governmental document intelligence, owning the full lifecycle from data curation and model development through API engineering, containerization, optimization and private on-premise deployment.

Deep expertise in VLMs, 4-bit QLoRA/PEFT, Unsloth, GGUF quantization, llama.cpp, vLLM, Ollama, YOLOv8, DINOv2, OpenCV, RapidOCR, PyMuPDF, Azure AI Document Intelligence, Google Gemini, FastAPI, Python concurrency, FAISS, SentenceTransformers, RapidFuzz and Arabic-English document processing. Delivered ~90% reduction in manual processing time, 70–80%+ lower cloud OCR expenditure, underwriting document review cut from ~3 hours to under 10 seconds, 96.4% document classification accuracy.

TECHNICAL SKILLS

- **Artificial Intelligence:** Vision-Language Models (VLMs), Large Language Models (LLMs), Small Language Models (SLMs), Generative AI, PEFT/LoRA/QLoRA, GGUF, NLP, Semantic Search.
- **Machine Learning:** Linear Regression, Logistic Regression, Decision Tree, Random Forest, KNN, K-Means Clustering.
- **Deep Learning:** CNN, LSTM, BERT
- **Frameworks:** PyTorch, Tensor Flow, Hugging Face Transformers, PEFT, Unsloth, Scikit-learn, Langchain
- **Foundation / Generative Models:** Qwen3.5-122B, Qwen3-VL, Gemma 4 E2B, Llama 4 Scout, Google Gemini
- **Document AI / OCR:** Intelligent Document Processing (IDP),Arabic OCR, Key Information Extraction (KIE), Layout Analysis, Document Classification, PDF Forensics, OCR Post-Processing, PyMuPDF, RapidOCR, PaddleOCR, Tesseract, PyPDFium2, Azure AI Document Intelligence (custom extraction neural, prebuilt-layout, Query Fields), Google Cloud Vision
- **Computer Vision:** Ultralytics YOLOv8, Meta DINOv2, OpenCV, DeepFace, ArcFace, Object Detection, Image Segmentation, Facial Recognition, Perceptual Hashing (pHash, DCT), Image Quality Assessment.
- **Model Serving / MLOps:** Edge AI, vLLM, llama.cpp, Ollama, GGUF, Model Deployment, Human-in-the-Loop (HITL), Dataset Auditing, Model Validation
- **Backend / APIs:** Python 3.10/3.11, FastAPI, REST APIs, Uvicorn, Pydantic, AsyncIO, ThreadPoolExecutor, OpenAPI/Swagger, JSON Schema, Error Handling
- **NLP / Search / Data:** SentenceTransformers, FAISS, RapidFuzz, Entity Resolution, Arabic Reshaping, Bidirectional Text Processing, Pandas, NumPy, JSONL, Parquet, Data Validation, SQL.
- **Cloud / Infrastructure:** Microsoft Azure AI, Azure AI Translation, Google Gemini API, Google GenAI SDK, Docker, Linux, Ubuntu, Nginx, Hugging Face Hub, AWS
- **UI / Applications:** Streamlit, PyQt5, Jinja2, HTML5 Canvas, JavaScript
- **Domain:** InsurTech, BFSI, Motor Claims Automation, Claims Processing, Subrogation, Commercial Property & Casualty (P&C) Underwriting, KYC/AML, Identity Verification, Insurance Fraud Detection, Government / GCC Document Intelligence, Data Sovereignty.

KEY ACHIEVEMENTS

- Architected and deployed 5 production AI systems and 7 POC projects across enterprise insurance and document-intelligence workflows.
- Distilled a 122B-parameter multimodal teacher model into a compact edge-deployable VLM; built private on-premise multimodal inference with 4-bit GGUF models and llama.cpp.
- Curated and validated 960 regional document records through a human-in-the-loop data-quality workflow.
- Reduced cloud OCR/API expenditure by ~70–80%+ through local extraction and hybrid routing; reduced motor-claim processing/intimation effort by ~90%.
- Cut commercial policy clause-review time from ~3 hours to under 10 seconds; achieved 96.4% document classification accuracy with DINOv2 visual representations.
- Solved complex Arabic-English document-processing problems: RTL text, glyph shaping, spatial ordering and multilingual entity resolution.

PROFESSIONAL EXPERIENCE

Qatar Insurance Group (QIC) / Anoud Technologies

AI/ML Systems Architect / Lead Document Intelligence Engineer **Jan 2023 - Present**

- Architected and deployed 5 production-grade AI systems across motor claims, KYC/AML, underwriting, fraud detection, identity verification, OCR and GCC document intelligence.
- Designed end-to-end Document AI architectures integrating OCR, computer vision, deep learning, cloud AI and local SLMs/VLMs with deterministic extraction and post-processing.
- Built hybrid cloud-edge inference pipelines, reducing cloud API costs by 70–80%+ while improving processing latency.
- Deployed private multimodal AI solutions using 4-bit quantization, GGUF, llama.cpp, vLLM and Ollama for secure, on-premise document processing.
- Developed Arabic-English document intelligence solutions covering glyph shaping, RTL text reconstruction, spatial OCR, multilingual entity resolution and government document layouts.
- Built Streamlit-based HITL annotation and validation platforms with reviewer provenance, data integrity and fine-tuning dataset generation.
- Diagnosed complex AI and production issues involving VLM memorization, projector incompatibility, ONNX Runtime contention, concurrency deadlocks and PDF spatial extraction.

Amazon Development Center

Data scientist / Support Engineer **Feb 2021 - Dec 2022**

- Developed Generative AI and RAG solutions using LLMs, vector embeddings and knowledge graphs for document understanding, contextual Q&A, summarization, recommendations and conversational intelligence.
- Built multimodal, real-time AI assistants and chatbots processing text, tables and images with accurate, citation-backed responses.
- Engineered end-to-end GenAI pipelines covering document ingestion, indexing, retrieval and inference, with cloud deployment, CI/CD and monitoring.

PROJECTS

1. **Qatar Document Intelligence API**: Architected a hybrid FastAPI document extraction platform combining PyMuPDF and Azure AI Document Intelligence with Arabic text reconstruction and structured JSON validation
2. **Multimodal KYC Intelligence**: Architected an AI-powered KYC platform integrating OCR, fuzzy matching, image quality assessment, and ArcFace-based facial similarity verification.
3. **VLM Document Intelligence**: Architected a multi-provider document AI platform integrating YOLOv8 segmentation, two-stage VLM extraction, dynamic JSON schemas, and specialized GCC police report parsers.
4. **DINOv2 Document Classification**: Implemented a deep learning document classification pipeline using DINOv2 vision transformer embeddings, supervised classification, and confidence-based predictions.
5. **Multilingual Voice Intelligence**: Developed a Gemini-powered conversational claims extraction platform using multimodal audio processing, context retention, and automated collection of 13 structured claim attributes.

EDUCATION

Bachelor of Engineering - St Joseph college of engineering **July 2017 - June 2021**

Master of Business Administration - University of Madras **July 2021 - July 2023**